

DS Unit 4 Question Bank — Graphs & Graph Algorithms

Data Structures (DS) · Unit 4 · Graphs & Graph Algorithms · 2025

Important and frequently asked questions, grouped by marks.

Part A — Short answer (2 marks)

1. Define Graph Representations. Derive or justify the key properties involved.
2. Illustrate Breadth-First Search with a worked example.
3. Define Depth-First Search. Derive or justify the key properties involved.
4. What is meant by Minimum Spanning Trees? Discuss its advantages and limitations.
5. Write short notes on Dijkstra's Algorithm.
6. Write short notes on Topological Sorting.

Part B — Long answer (10 marks)

1. Define Graph Representations. Derive or justify the key properties involved.
2. Illustrate Breadth-First Search with a worked example.
3. Define Depth-First Search. Derive or justify the key properties involved.
4. Write short notes on Minimum Spanning Trees.
5. Write short notes on Dijkstra's Algorithm.
6. Discuss the role of Topological Sorting in Graphs & Graph Algorithms.
7. Discuss the role of Graph Representations in Graphs & Graph Algorithms.
8. Discuss the role of Breadth-First Search in Graphs & Graph Algorithms.
9. Discuss the role of Depth-First Search in Graphs & Graph Algorithms.
10. Explain Minimum Spanning Trees with a neat diagram and a suitable example.
11. Compare Dijkstra's Algorithm with Depth-First Search. When would you prefer one over the other?
12. Illustrate Topological Sorting with a worked example.